

Chapter 1 Jacobson Radical of Non-commutative Rings

Definition 1.0.0.1 (Jacobson Radical) Define: The Jacobson radical

$$\text{rad}(R) = \cap_{I \in L} I \quad L = \{I \subseteq R \mid I \text{ is the left ideal of } R\}$$

◇ **Lemma 1.0.0.2** For $y \in R$, the following statements are equivalent:

- $y \in \text{rad}(R)$;
- $1 - xy$ is left-invertible for any $x \in R$;
- $yM = 0$ for any simple left R -module M .

◦ **Proof**

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Annihilators

Definition 1.0.0.3 ...

Jacobson Semisimple

Nilpotent

1.1 Jacobson Radical Under Change of Rings